

Sound change and naturalness

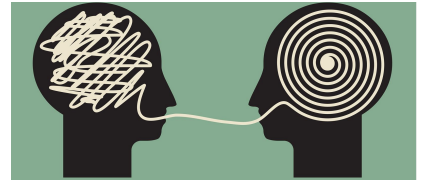
Ling 105: The Sound of Language
4th April 2023

what is “natural”?

so far: naturalness of phoneme inventories & phonological rules

today: naturalness of historical sound changes

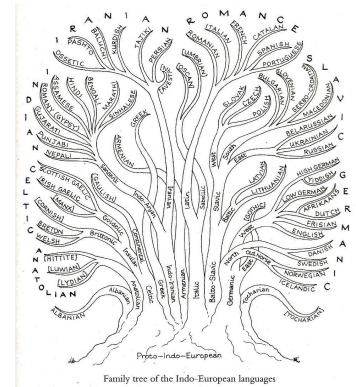
- typologically frequent (found in many of the world’s languages)
- phonetically motivated
 - articulatory: minimize effort
 - perceptual: maximize distinctness
- ease of learning (in child and adult language acquisition)



some typologically frequent sound changes

- **assimilation**: a sound becomes more like another sound
- **palatalization**: a consonant becomes more palatal
- **metathesis**: two sounds switch places
- **epenthesis**: a sound is inserted between two adjacent sounds
- **deletion** (**elision**): a sound is deleted
- **lenition** (**weakening**): a sound becomes softer or less strong
- **tonogenesis**: tonal contrasts emerge from non-tonal contrasts

do these sound changes sound familiar? where have we seen them before?



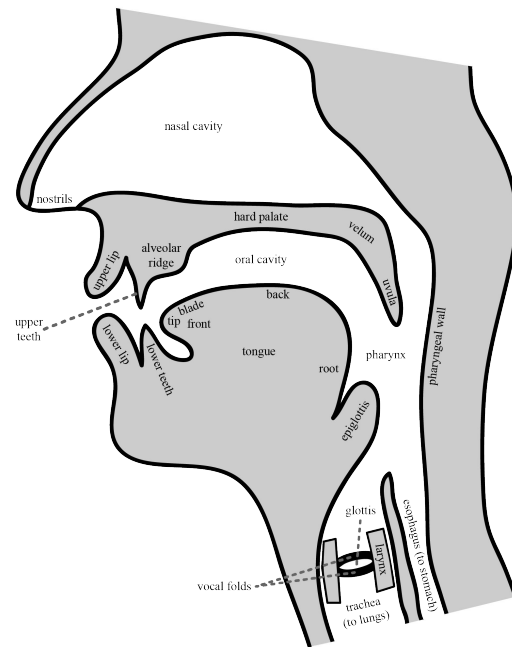
some typologically frequent sound changes: examples

- **assimilation**, e.g. nasal assimilation in *contact* vs. *companion*
- **palatalization**, e.g. English *sugar* ['ʃʊgə] vs. Spanish *sucré* ['sukre]
- **metathesis**, e.g. *ask* ['æsk] vs. ['æks] from Old English *ascian* vs. *acsian*
- **epenthesis**, e.g. French *passager* > English *passenger*
- **deletion** or **elision**, e.g. *opossum* > *possum*
- **lenition** or **weakening**, e.g. English *father* ['fɑðə] vs. Latin *pater* ['pater]
- **tonogenesis**, e.g. Korean plain vs. aspirated consonants > L vs. H tone

do these sound changes sound familiar? where have we seen them before?

phonetic motivations for sound change

- which phonological change is more natural?
 - a. $V \rightarrow [+back] / [+back] C _$
 - b. $V \rightarrow [+back] / [+high] C _$
- which phoneme inventory is more natural?
 - a. /p t b d/
 - b. /t̪ ɬ d ɖ/



we can think of (diachronic) sound changes as (synchronic) phonological changes that have solidified over time, such that their motivations may no longer be transparent

phonetic motivations for sound change

- articulatory motivations: minimize effort
- perceptual motivations : maximize distinctness

question: why don't languages gradually become more simple or more complex? how do these forces work against each other?

testing ease of learning

1. due language specific cognitive faculty, [Universal Grammar \(UG\)](#)
 2. due to general cognitive functioning
- in artificial grammar learning experiments
 - English speakers learn H (high tone) spreading more easily than L (low tone) spreading (Glenn 2012)
 - more natural: H spreads rightwards: $/H-L/ \rightarrow /H-H/$
 $/L-H/ \rightarrow /L-H/$
 - less natural: L spreads rightwards: $/H-L/ \rightarrow /H-L/$
 $/L-H/ \rightarrow /L-L/$

testing ease of learning

Part V: In this section, you will be asked to assess two versions of one word. Press the "Play versions 1 & 2" button to begin. Listen carefully to both versions. You must press the play button twice. After hearing each version twice, you will be asked to choose which version you thought sounded better by clicking either "Version 1 sounded better" or "Version 2 sounded better." You may press the "Proceed to next word" button to continue.

You must listen twice.

Play versions 1 & 2

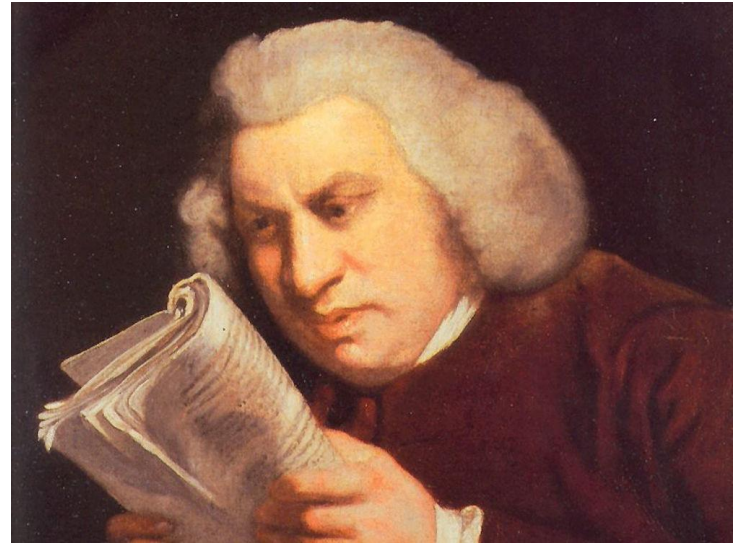


Version 1 sounded better

Proceed to next word

Version 2 sounded better

let's look at some case studies...



Spanish lenition

1. **spirantization**: voiced oral stops become fricatives between vowels
2. **debuccalization**: syllable-final /s/ loses its place of articulation as [h]
in some dialects, [es'tados u'nidos] becomes [eh'tadoh u'nido]

[ˈbaka]	“cow”
[la ˈβaka]	“the cow”
[ˈgwapa]	“pretty one”
[la ˈɣwapa]	“the pretty one”
[deˈretʃa]	“right”
[la ðeˈretʃa]	“the right”
[ˈtarde]	“evening”
[la ˈtarde]	“the evening”

tip #1: intervocalic contexts V_V are a common environment that leads to sound change cross-linguistically

- compare to English flapping
e.g. **write** [ˈɹaɪt]
writer [ˈɹaɪrə]
ride [ˈɹaɪd]
rider [ˈɹaɪrə]

Seoul Korean tonogenesis

1. “plain” aspirated stops: leads to low (L) tone on the syllable
2. “aspirated” aspirated stops: leads to high (H) tone on the syllable
3. “tense” unaspirated stops: does not partake in tonal contrast

['pʰùl]	“fire”
['pʰúl]	“grass”
['tàsán]	“fertility”
['tásán]	“calculation”
[kʰù'lim]	“painting”
[kʰú'lim]	“cream”

tip #2: differences in voice onset time (VOT) can lead to tonal contrasts, where longer VOT corresponds to higher pitch

- compare to tonogenesis in Chinese:
[https://en.wikipedia.org/wiki/Four_tones_\(Middle_Chinese\)](https://en.wikipedia.org/wiki/Four_tones_(Middle_Chinese))
- what do you think are the different origins of level vs. contour tone?

Grimm's Law

English

foot [f]

three [θ]

ten [t]

cold [k]

father [f]

hound [h]

Spanish

pie

tres

diez

—

padre

can

Greek

podós

treîs

déka

gelandrós

patér

kýōn

Lithuanian

pėda

trīs

dešimt

gelmenis

—

—

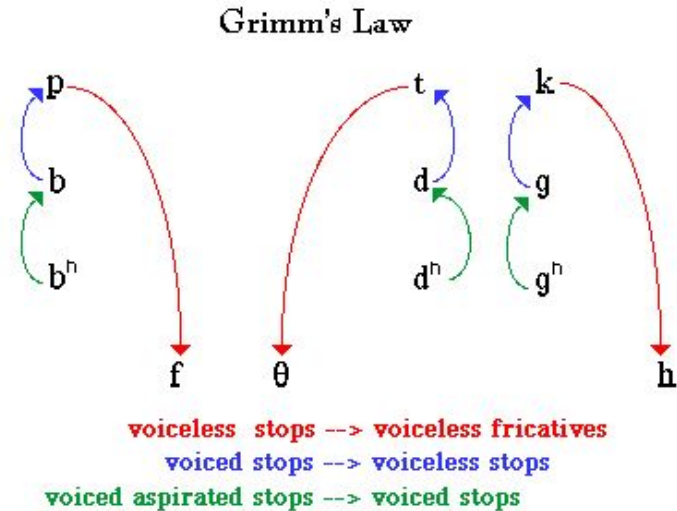
Grimm's Law

Germanic stop consonants as developed from Proto-Indo-European (PIE)

1. aspirated voiced stops became voiced stops
2. voiced stops became voiceless stops
3. voiceless stops became voiceless fricatives

tip #3: chain shifts, or a set of sound changes that are linked and could be thought of in a “push” or “pull” sense, are a common diachronic sound change

- compare English Great Vowel Shift



California / Northern Cities / Canadian vowel shift



California / Northern Cities / Canadian vowel shift

the **vowel space is becoming more centralized**, with notable consequences:

<https://www.youtube.com/watch?v=dCer2e0t8r8>

how do you pronounce the following?

- *cot* vs. *caught*, *bot* vs. *bought*
- *marry* vs. *Mary* vs. *merry*
- *kettle* vs. *cattle*
- *book* vs. *buck*
- *deck* vs. *duck*, *wet* vs. *what*
- *true* vs. *tree*, *rude* vs. *reed*

finish this sentence: *you get what you get...*

